



Shifting Stories, Shifting Returns

Team: Eigenvector

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Motivation & Relevance

Understanding Narrative Shifts

Exploring how changes in corporate narratives reflect evolving business outlooks, and underlying fundamentals

Translating Signals into Stories

Transforming linguistic and contextual shifts in financial reports into measurable, predictive investment signals



Bridging Text and Market Data

Connecting qualitative disclosures with quantitative market behavior to uncover hidden drivers of stock performance

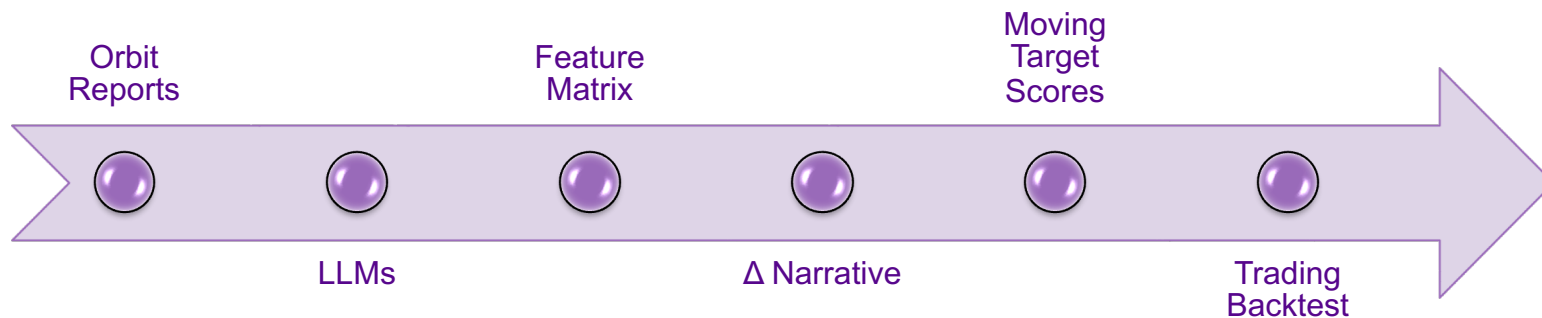
Interpreting Real-World Impact

Revealing how corporate storytelling anticipates market reactions, bridging academic insights with practical relevance

Firms don't change what they do — they change how they talk about it

Inspired by *Cohen & Nguyen (2024, "Moving Targets")*, which shows narrative focus shifts when fundamentals deteriorate.

Methodology Overview



Narrative Dimensions

01. Financials

- Sales, Expense, Profit, Operations, Liquidity, Investment, Financing, Litigation, Employment, Tax, Accounting

Fengler & Phan (2023), "A Topic Model for 10-K Management Disclosures"

02. Business

- Product, Buyer, Process, Location, Promotion, Supplier, Management, Strategy, Industry, Business description

Beattie, McInnes, & Fearnley (2004), "A Methodology for Analysing and Evaluating Narratives in Annual Reports". Accounting Forum, 28(3), 205–236

03. Risk

- Firm specific vs Generic boilerplate

Huang & Li (2008), "A Multilabel Text Classification Algorithm for Labeling Risk Factors in SEC Form 10-K", ACM TMIS

04. Timeline

- Forward looking vs Historical

Muslu et al. (2015), "Forward-Looking MD&A Disclosures and the Information Environment," Management Science, 61(5), 931–948

05. Sentence Structure

- Subjective vs Objective

Kogan & Meursault (2022), "Corporate Disclosure: Facts or Opinions?", FRB of Philadelphia

06. Sentiment (Tone)

- Positive / Neutral / Negative

Zeng & Jiang (2023), "Financial Sentiment Analysis Using FinBERT with Application in Predicting Stock Movement"

07. Analyst

- If a sentence originated from an analyst

Vanhaverbeke et al. (2024), "Mandatory Financial Information Disclosure and Credit Ratings," Journal of Accounting and Economics, 78(1)

Initial LLM Prompt Approach

```
prompt = f"""
Classify the following sentence into categories.
Each category should be 0 (not present) or 1 (present).
Sentiment can be -1 (negative), 0 (neutral), or +1 (positive).

Base your assessment only on the documents provided and not your training knowledge of the company, industry or market.

Categories:
Financial: Sales/Revenue, Cost/Expense, Profit/Loss, Operations, Liquidity, Investment, Financing, Litigation, Employment, Regulation/Tax,
Accounting
Non-financial: Product, People (buyer), Process, Location, Macro environment, Promotion, People (seller), Management
Risk related: Firm specific risk, Generic boilerplate
Objective vs Subjective: Objective, Subjective
Timeframe: Forward looking, Historical
Sentiment Score: -1, 0, +1

Return JSON like:
{{
  "Sales/Revenue": 1, "Cost/Expense": 0, ..., "Sentiment": +1
}}

Sentence: "{sentence}"
"""
```

Among open-source models,
GPT-OSS-120B performs best
but lacks continuous score (eg.,
0.4, 0.7) to measure the
narrative intensity and overlap

Output

```
Model: gpt-oss-120b
{'Sales/Revenue': 0, 'Cost/Expense': 0, 'Profit/Loss': 0, 'Operations': 0, 'Liquidity': 1, 'Investment': 1, 'Financing': 0, 'Litigation': 0,
'Employment': 0, 'Regulation/Tax': 0, 'Accounting': 0, 'Product': 0, 'People (buyer)': 0, 'Process': 0, 'Location': 1, 'Macro environment':
0, 'Promotion': 0, 'People (seller)': 0, 'Management': 0, 'Firm specific risk': 0, 'Generic boilerplate': 1, 'Objective': 1, 'Subjective': 0,
'Forward looking': 1, 'Historical': 1, 'Sentiment': 0}
```

Initial LLM Prompt Approach

Asking LLM to give continuous scores

📘 Sentence:

Capital Spending—Capital expenditures were \$44.6 million and \$26.2 million for the six months ended June 30, 2024 and 2023, respectively, and primarily related to the purchase of equipment to increase our manufacturing capacity. We expect capital expenditures for 2024 to increase compared with 2023 given the timing of spending on machinery, equipment and tooling for our new Malaysia manufacturing facility and to support continuous improvement efforts in our other manufacturing locations. To a lesser extent, we expect capital expenditures to increase due to investments in our information technology infrastructure. We expect to fund our capital expenditures using existing cash.

=====

🤖 Model: openai-120b

```
{  
  "Sales/Revenue": 0.0,  
  "Expense/Cost": 0.15,  
  "Profit/Loss": 0.0,  
  "Operations": 0.4,  
  "Liquidity": 0.5,  
  "Investment": 0.9,  
  "Financing": 0.3,  
  "Litigation": 0.0,  
  "Employment": 0.0,  
  "Regulation/Tax": 0.0,  
  "Accounting": 0.1,  
  "Capital Spending": 0.15  
}
```

Even when the sentence clearly describes **capital spending**, the model assigns only **0.15** to **Expense/Cost**

Token Probabilities $\neq$ Semantic Similarity

Embedding-based LLM Approach

- Compute **cosine similarity** with representative words of each category

```
"Sales/Revenue": [  
    "sales", "revenue", "turnover", "top line", "net sales", "gross sales",  
    "service income", "product revenue", "subscription revenue", "customer receipts",  
    "contract revenue", "retail revenue", "distribution income", "sales growth",  
    "sales volume", "sales pipeline"  
],  
"Expense/Cost": [  
    "operating expenses", "cost of goods sold", "COGS", "administrative expenses",  
    "selling expenses", "marketing spend", "R&D expense", "depreciation expense",  
    "amortization", "general and administrative costs", "overheads", "freight cost",  
    "utilities expense", "rent expense", "interest expense", "insurance expense"  
],
```

- Any word's **similarity** > **0.35**, assign that value as the category score; otherwise 0

```
=== Finance Category Similarities ===  
Sales/Revenue      | avg: 0.236 | max: 0.313  
Expense/Cost       | avg: 0.286 | max: 0.365 ✓ (above threshold)  
Profit/Loss        | avg: 0.239 | max: 0.297  
Operations         | avg: 0.274 | max: 0.421 ✓ (above threshold)  
Liquidity          | avg: 0.282 | max: 0.371 ✓ (above threshold)  
Investment         | avg: 0.331 | max: 0.536 ✓ (above threshold)  
Financing          | avg: 0.269 | max: 0.410 ✓ (above threshold)  
Litigation         | avg: 0.123 | max: 0.226  
Employment         | avg: 0.193 | max: 0.278  
Regulation/Tax     | avg: 0.172 | max: 0.262  
Accounting         | avg: 0.233 | max: 0.336
```

Better LLM Approach



Financial & Business

Cosine Similarity

OpenAI text-embedding-3-small



Risk

Bart NLI



Timeline

(Forward vs. Historical)

Rule-based
using tense and verbs



Sentence Structure

(Objective vs. Subjective)

Rule-based



Analyst

Bart NLI



Sentiment

FinBERT

Data Insights

Bang-Bang Max Delta

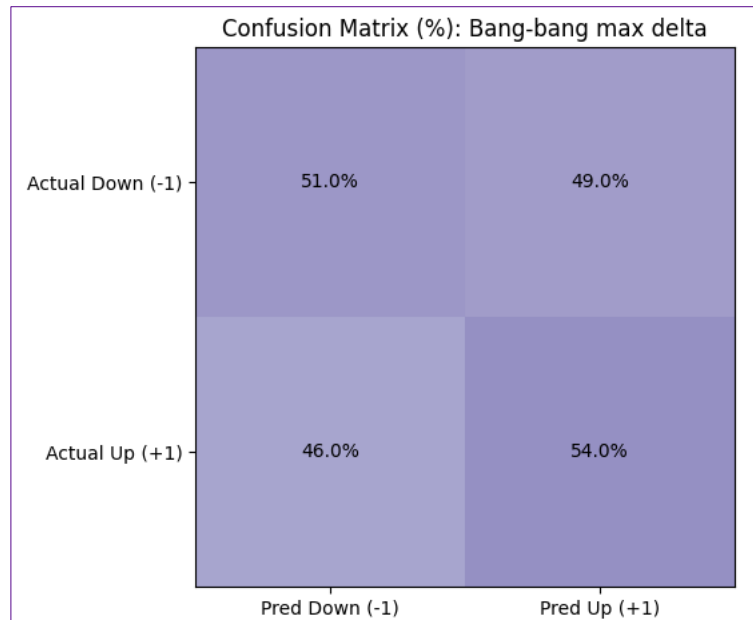
- Strongest narrative change ($\max |\Delta|$) carries strongest predictive information
- Achieves balanced directional accuracy ~54% Up and ~51% Down predictions

Distribution of Narrative Shifts

- Firms averaged +15 feature changes per quarter
- Corporate narratives are highly dynamic; subtle text changes are frequent

Financial vs Non-Financial Shifts

- ~90% of firm-quarters show changes in both financial and business topics
- Non-financial framing often precedes observable financial deterioration



Data Insights

Timeframe Orientation

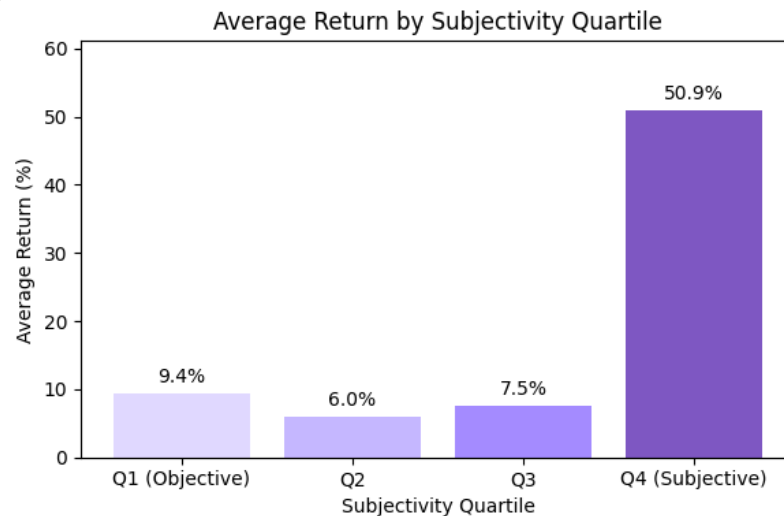
- Firms with higher future orientation earned 30.4% average returns versus 7.1% for less forward-looking firms.
- Strong focus on future prospects is associated with substantially higher subsequent performance

Sentence Structure (Subjective vs Objective)

- Narrative-driven, opinion-heavy language links to higher returns
- Subjective (top quartile) → ~50.8% returns;
Objective → ~9.4%

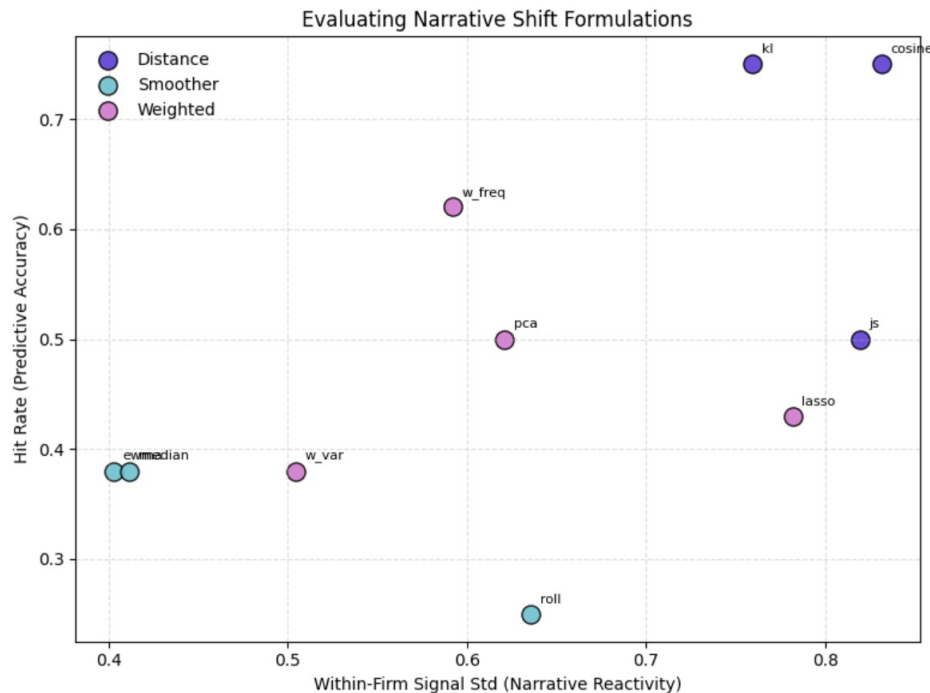
Risk Language Dynamics

- Generic risk disclosures correlated (~0.14) with subsequent returns, more strongly than any other category
- Market reacts more to risk tone than idiosyncratic warnings



Quantifying Narrative Shift

- Tested multiple MTS formulations (rolling, weighted, distance-based)
- **Smoothers** (Roll, EWMA, Median): Gradual trend capture → missed abrupt shifts.
- **Weight-based** (Var, Freq, PCA, Lasso): High noise, unstable weighting, weak signal.
- **Distance-based** (KL, Cosine): Captured proportional & directional changes — most robust.



Quantifying Narrative Shift

MTS KL

- Measures how the distribution of narrative categories changes in relative importance.
- **Large KL** → management heavily reframed priorities → potential underlying stress or pivot.
- **Small KL** → consistent emphasis across categories → stable business outlook

$$MTS_{KL}(t) = \sum_{i=1}^N p_{i,t} \log \left(\frac{p_{i,t}}{p_{i,t-1}} \right),$$

Where:

- $p_{i,t}$ = normalized weight of category i in quarter t
- $p_{i,t-1}$ = normalized weight of category i in previous quarter
- N = total narrative categories
- All p-values sum to 1 each quarter.

MTS Cosine

- Looks at alignment between current and previous quarter narratives in feature space.
- **Large Cosine** → narrative direction flipped or reversed → signals abrupt change in communication focus or managerial stance.
- **Small Cosine** → consistent evolution → steady continuation of prior messaging.

$$MTS_{\cos}(t) = 1 - \frac{\sum_{i=1}^N f_{i,t} f_{i,t-1}}{\sqrt{\sum_{i=1}^N f_{i,t}^2} \sqrt{\sum_{i=1}^N f_{i,t-1}^2}}$$

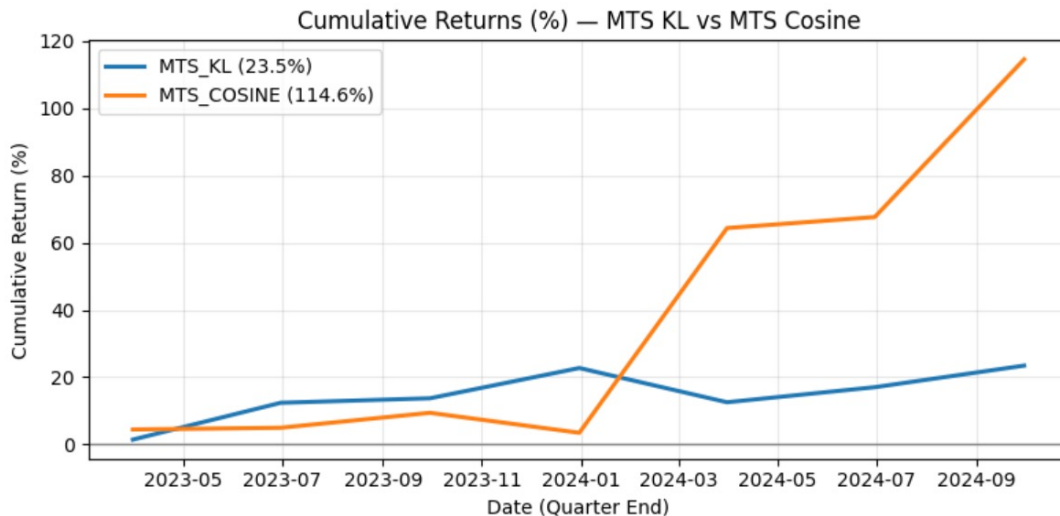
Where:

- $f_{i,t}$ = feature value of category i in quarter t
- $f_{i,t-1}$ = feature value of category i in previous quarter
- N = total narrative categories
- Each f-vector represents the weighted narrative profile per quarter.

Trading Strategy

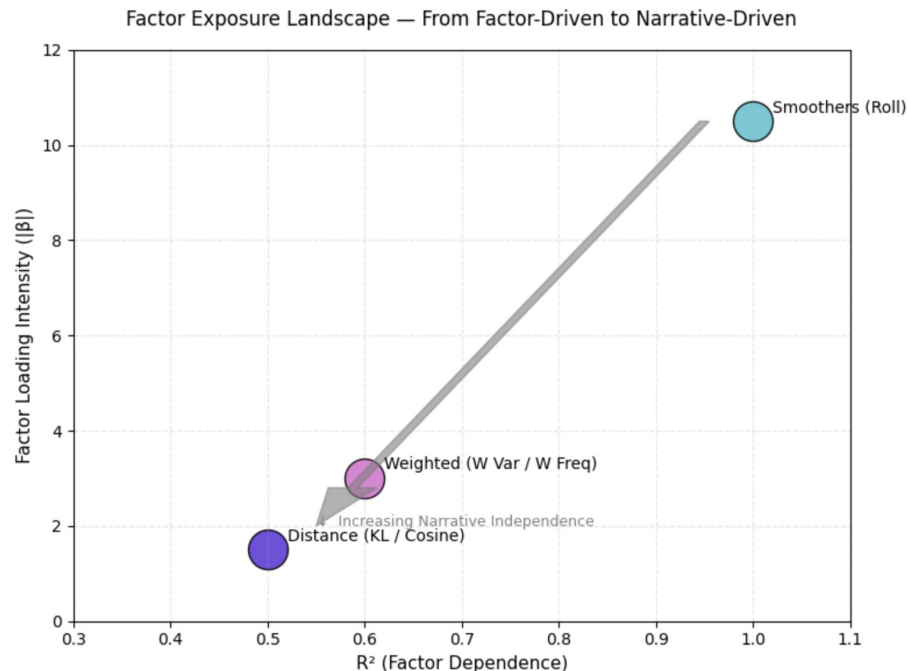
- **Entry:** On the first trading day of the month following each firm's report release.
- **Exit:** After the first month following the next quarterly report ($\approx$ one-quarter holding period).
- **Positioning:** 10–10% flipped long–short; short high-MTS (unstable narratives), long low-MTS (stable narratives).

MTS Method	Mean	Sharpe	Hit Rate
MTS roll	−8.37	−0.36	0.25
MTS w var	−0.01	−0.16	0.38
MTS w freq	−0.06	−0.29	0.62
MTS KL	+0.03	0.46	0.75
MTS Cosine	+0.12	0.54	0.75



Factor-Controlled Analysis

- **Smoothers (Roll)** show $R^2 \approx 1.0$ with **extreme factor betas** → returns fully explained by known styles.
- **Weighted variants** (W Var / W Freq) show moderate exposures but **no consistent alpha** after controlling for market and momentum risk.
- **Distance-based signals** (KL, Cosine) maintain **lower R^2 (≈ 0.5 – 0.6)** and moderate betas, suggesting orthogonality to standard factors.
- **KL and Cosine** remain most interpretable and resilient: their predictive edge likely reflects **narrative-driven information** rather than factor tilts.



Risk Management & Pitfalls

LLM Classification Bias

- Model can misclassify nuanced phrases
- **Mitigation** → combine LLM labels with rule-based filters and embedding thresholds

Factor Exposure Overlap

- Narrative scores may load on momentum/value factors → false alpha
- **Mitigation** → run quarterly factor regressions with HAC errors

Statistical Stability & Sample Length

- Quarter window limits t-stat power; alpha estimates unstable
- **Mitigation** → bootstrap portfolios and extend horizon for out-of-sample validation

Interpretability & Overfitting Risk

- High-dimensional feature space can capture noise as signal.
- **Mitigation** → prefer distance-based measures (KL, Cosine) for structure not content



NYU

Thank You